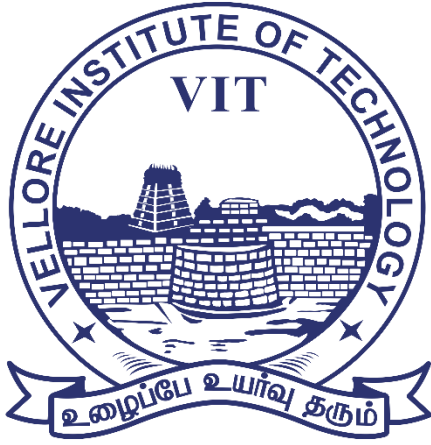




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Fall Semester 2025-26

Project Report

Project Title: Scientific Calculator

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Course: Fundamentals in AI and ML.

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Class Number: BL2025260500819

Slot: B11+B12+B13

Multifunctional Calculator

Course: Fundamentals of Artificial Intelligence and Machine Learning

1. Introduction

In the modern digital world, calculators have evolved beyond simple arithmetic tools into multifunctional systems capable of performing complex scientific and analytical tasks. This project, titled “**Multifunctional Calculator**”, is a Python-based application designed to integrate algebraic, scientific, and data visualization functionalities into a single system.

The project demonstrates how computational tools can mimic basic intelligent decision-making processes by interpreting user input and dynamically selecting appropriate operations. This connects directly with foundational concepts in **Artificial Intelligence (AI)** such as user interaction, decision logic, and automation.

2. Problem Statement

Traditional calculators are limited in functionality and often require multiple tools for different types of operations such as:

- Basic arithmetic
- Scientific computations
- Data visualization

There is a need for a unified system that:

- Combines multiple functionalities
- Provides an interactive user experience
- Demonstrates basic AI-driven decision-making

3. Objectives

- To develop a **multifunctional calculator** using Python
- To integrate **algebraic, scientific, and graphical operations**
- To implement **basic AI concepts** such as input interpretation and decision-making
- To enhance usability through interactive command-line input
- To demonstrate structured programming and modular design

4. Relevance to AI & ML Concepts

Although this is not a full ML model, the project incorporates **basic AI principles**, which are appropriate for a first-year course:

Intelligent Decision Making

- The program uses conditional logic (if-elif) to **interpret user input** and select the correct operation.
- This mimics a simple **rule-based AI system**.

Human-Computer Interaction

- The system interacts with the user through prompts and responses.
- It processes natural-like commands such as "addition", "sine", etc.

Automation

- Once input is provided, the system automatically executes the required task without further intervention.

Data Handling & Visualization

- The use of graphs (line, bar, pie) introduces basic concepts of **data representation**, which is important in ML.

Conclusion (AI relevance):

The calculator behaves like a **basic intelligent agent** that takes input, processes it using predefined logic, and produces output.

5. System Description

The calculator is divided into three main modules:

1. Algebraic Operations

- Addition
- Subtraction
- Multiplication
- Division
- Random Number Generator

2. Scientific Operations

- Factorial
- Fibonacci Series
- Exponential (e^x)
- Logarithm
- Trigonometric Functions:
 - Sine
 - Cosine
 - Tangent
 - Cosecant
 - Secant
 - Cotangent

3. Data Visualization

- Line Graph
- Bar Graph
- Pie Chart

6. Methodology

1. User Input Phase

- User enters their name and selects a function.

2. Input Validation

- The system checks if the input matches available functions.

3. Decision Logic

- Uses if-elif statements to determine the operation.

4. Execution

- Performs calculations using:
 - math
 - numpy
 - random
 - matplotlib

5. Output Display

- Displays numerical results or graphs.

7. Tools and Technologies Used

- **Programming Language:** Python

- **Libraries Used:**

- math → Scientific calculations
- numpy → Numerical operations
- random → Random number generation
- matplotlib → Data visualization

8. Code Organization and Documentation

Strengths:

- Logical separation of functionalities
- Use of descriptive prompts
- Clear input-output structure

Areas of Improvement:

- Code can be modularized into functions
- Repeated conditions can be optimized
- Comments can be added for better readability

9. Results and Output

The calculator successfully:

- Performs arithmetic and scientific calculations
- Generates correct trigonometric values
- Displays graphs based on user input
- Produces random numbers within a given range

10. Limitations

- Case-sensitive input handling (limited flexibility)
- No GUI (only command-line interface)
- No error handling for invalid numeric inputs (e.g., division by zero)
- Not using advanced AI/ML models

11. Future Enhancements

- Add **Natural Language Processing (NLP)** for better input understanding
- Develop a **Graphical User Interface (GUI)**
- Integrate **machine learning** for predictive calculations
- Add voice input support
- Improve error handling

12. Reflection (Important for Evaluation)

This project helped in understanding how even simple programs can simulate **basic AI behavior**. By implementing decision-making using conditional statements, we realized how machines can process inputs and produce intelligent outputs.

The integration of multiple functionalities into a single system improved our understanding of:

- Problem-solving
- Code structuring
- User interaction

Additionally, working with visualization tools provided insight into how data is represented—an essential aspect of machine learning.

13. README (Usability Section)

How to Run the Project

1. Install Python
2. Install required libraries:
`pip install numpy matplotlib scipy`
3. Run the .ipynb file in Jupyter Notebook

How to Use

- Enter your name
- Choose a function (e.g., Addition, sine, Bar Graph)
- Provide required inputs
- View results or graphs

Example Commands

- addition
- sine
- bar graph

14. Conclusion

The **Multifunctional Calculator** successfully demonstrates how basic programming concepts can be combined with introductory AI principles to create an intelligent system. It is a practical example of how machines can interpret inputs, make decisions, and perform complex operations efficiently.

